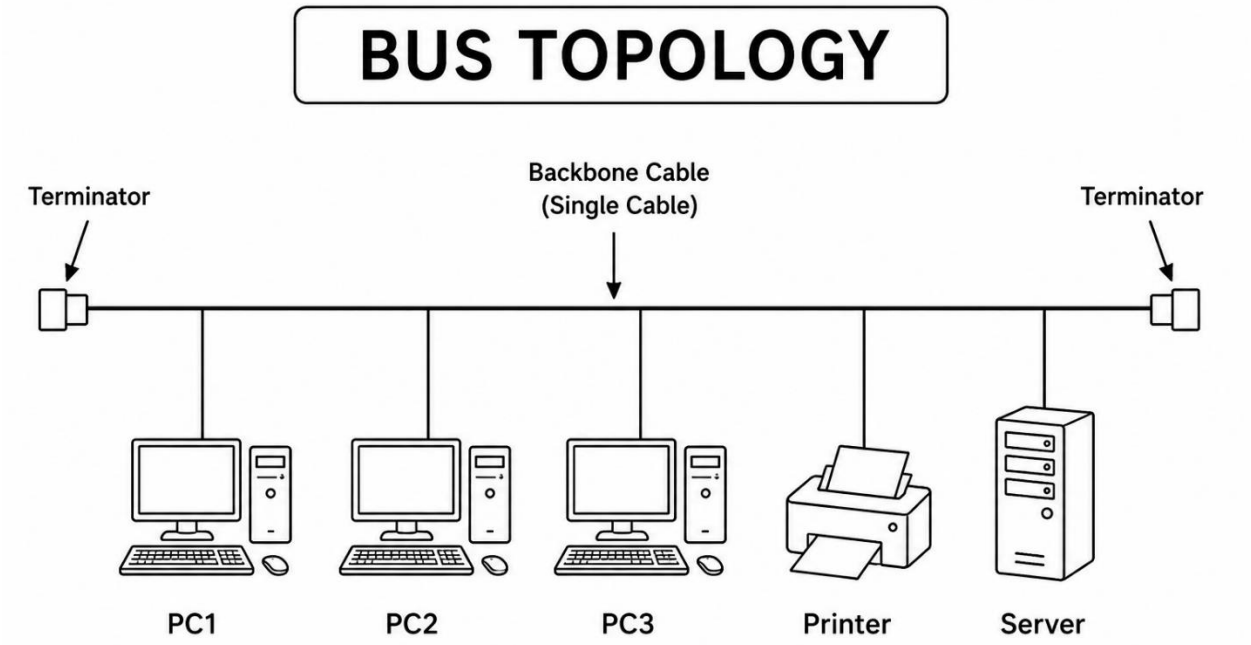
Day 2: Network Topologies Theory :Types of network topologies: Bus, Star, Ring, Mesh. Practical :Simulate a star topology using Cisco Packet Tracer. Task: Draw a diagram of topologies with examples.

Tasks:-

1. Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.

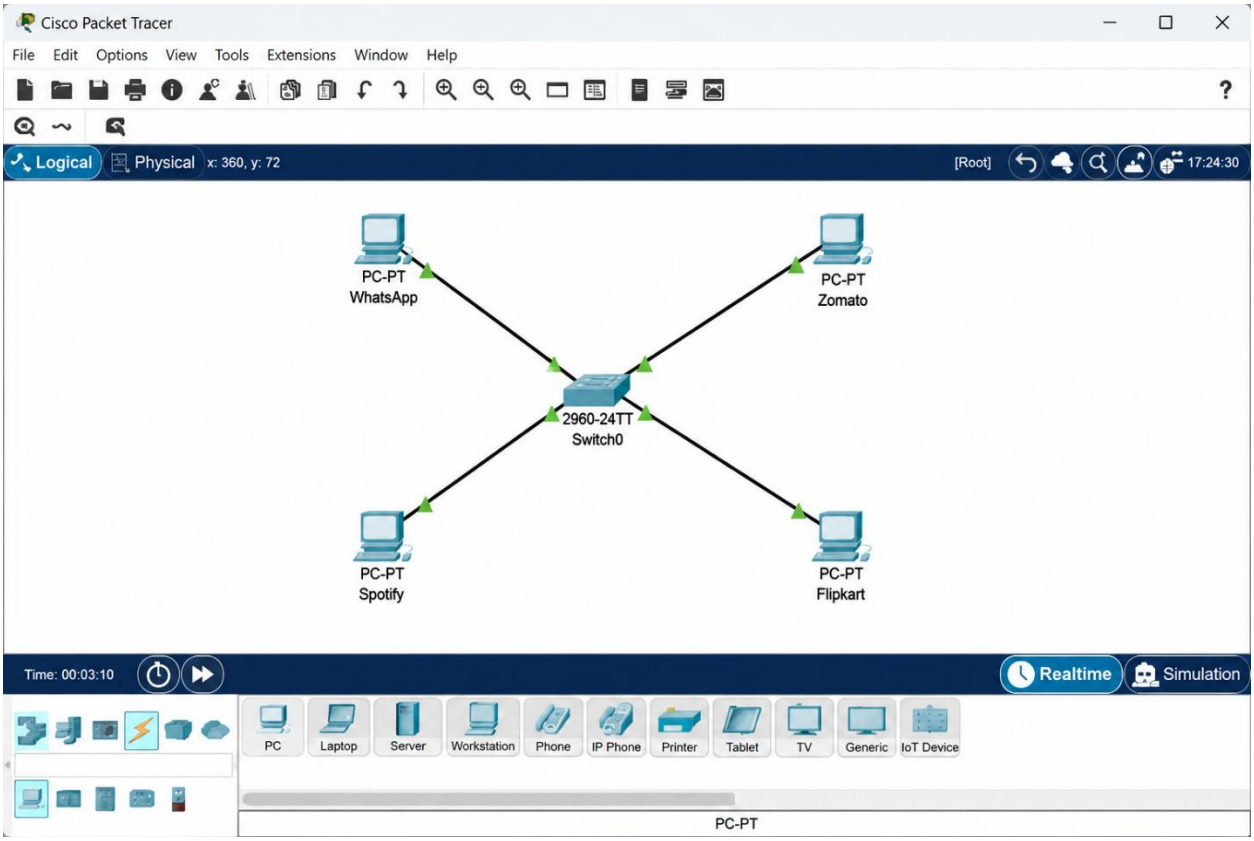
Ans.:-



A small office or computer lab with a limited number of computers can use a bus topology. All computers share a single backbone cable to communicate with each other and access shared resources such as a printer or file server. Although it is inexpensive and easy to install, it is rarely used today because a fault in the backbone cable can bring down the entire network.

2. In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.

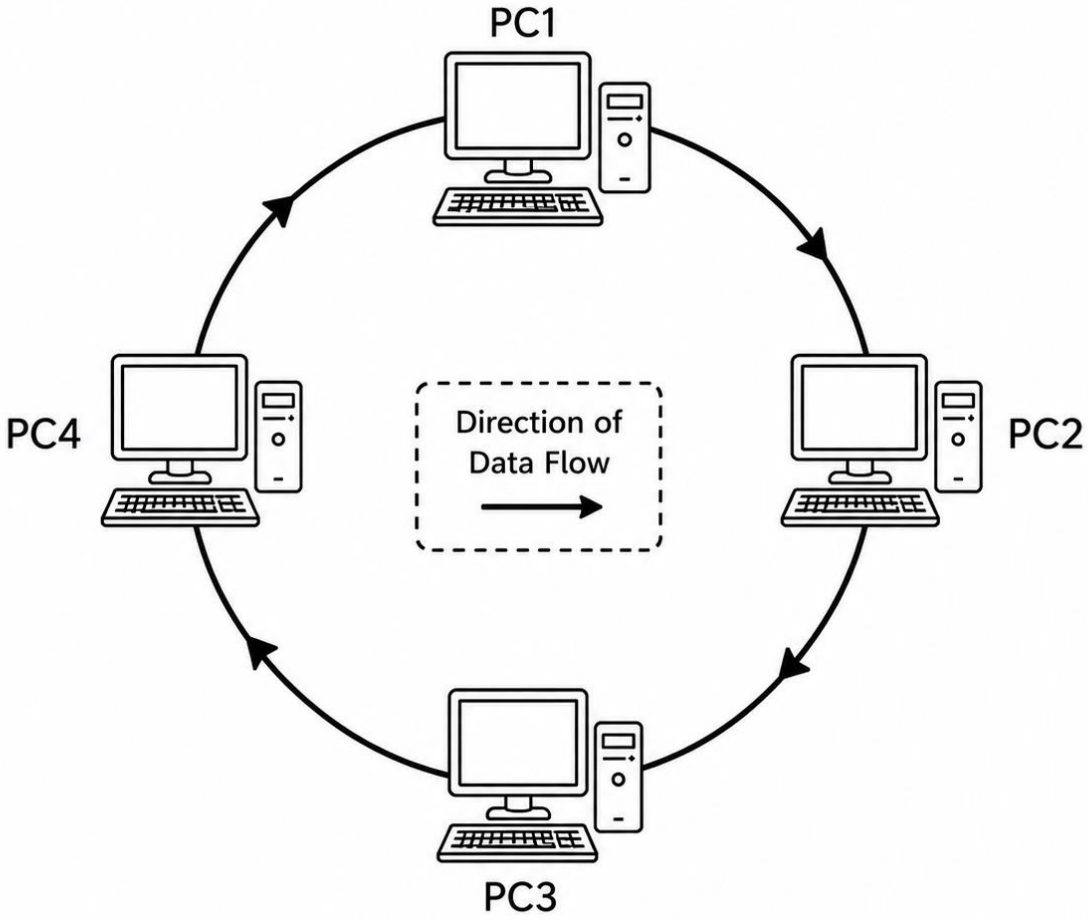
Ans.:-



**3. Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.

Hint: Use arrows to show the direction of data flow.**

Ans.:-



In a ring topology, data travels in one direction from one device to the next around the ring. Each device receives the data, checks the destination, and forwards it if needed. This continues until the data reaches the correct device. Since devices transmit one at a time, data collisions are minimized. If one device or cable fails, the network may stop working unless a backup ring is available.

4. Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

Ans.:-

Mesh Topologies:-

Advantages:-

1. High Reliability: Multiple paths keep the network running even if one connection fails.
2. Excellent Fault Tolerance: Failure of one device or cable does not affect the rest of the network.
3. Better Security and Performance: Dedicated links reduce congestion and make data transmission more secure.

Disadvantages:-

1. High Cost: Requires many cables and network ports.
2. Complex Installation: Setting up and maintaining the network is difficult.
3. Difficult to Expand: Adding a new device requires connections to all existing devices.

Star Topologies:-

Advantages:-

1.Easy to Install and Manage: Devices can be added or removed easily.
2. Good Performance: Each device has a dedicated connection to the central switch, reducing data collisions.
3. Easy Troubleshooting: Faults are easy to identify because each device is connected separately.

Disadvantages:-

1.Central Device Dependency: If the switch or hub fails, the entire network stops working.
2. Higher Cabling Cost: More cables are required than in a bus topology.
3. Limited by Switch Capacity: The number of devices depends on the available ports on the switch.